SSMD: Dimensionality Reduction and Classification of Hyperspectral Images Based on Spatial–Spectral Manifold Distance Metric Learning

Yao Jin, Yanni Dong, *Senior Member, IEEE*, Yuxiang Zhang, *Member, IEEE*, and Xiangyun Hu

***Abstract*— Metric learning, which aims to obtain a metric matrix M such that samples from the same class are close to one another and samples of different classes are far from one another, is widely used in the field of hyperspectral dimensionality reduction (DR) and classification. Traditional metric learning is based on the Mahalanobis distance, which measures the similarity between samples via point-to-point distance, ignoring the structural features of the hyperspectral images (HSIs). To solve the above problem, we proposed clustered multiple manifold metric learning (CM^3L), which obtains a manifold distance (MD), aimed at improving discrimination by introducing structural features of the HSIs and achieving good results. However, this manifold distance still has certain shortcomings in specific application situations. MD only considers the labeled data in the construction of the manifold and ignores the unlabeled data, resulting in the destruction of the manifold. Therefore, this article proposes a new spatial–spectral manifold distance (SSMD) to improve the performance of metric learning in hyperspectral DR and classification by maintaining the integrity of the constructed manifolds. The SSMD selects suitable neighboring points in the labeled and unlabeled data through the spectral–spatial information in order to participate in the construction of the manifold. Then, the distance between the manifolds is calculated to replace the traditional Mahalanobis distance. The results of seven sets of comparison experiments on three real HSI datasets demonstrate the effectiveness of SSMD in improving the classification results of HSIs.**

***Index Terms*— Dimensionality reduction (DR), hyperspectral image (HSI), manifold distance, metric learning.**

I. Introduction

HYPERSPECTRAL images (HSIs) have numerous spectral bands, which can acquire the spectral information of tens to hundreds of continuous bands of surface objects. HSIs also combine spectral information with spatial information, reflecting the texture, morphology, and other characteristics of surface objects, which has the characteristic of "map-spectrum integration" [1], [2], [3], [4]. Thus, HSIs are richer in spectral band information, effectively improving their ability to identify surface objects [5], [6], [7]. However, the large number of bands and their narrow ranges lead to a certain degree of similarity between adjacent bands, so HSIs have a large amount of redundant information and are prone to dimensional catastrophe, which limits the subsequent application of hyperspectral images (HSIs) [8], [9], [10]. Dimensionality reduction (DR) techniques, which are effective means to solve the above problems, can find a low-dimensional subspace and generate new features that better express the essence of the original data while eliminating redundant data [11], [12], [13].

Hyperspectral DR techniques, which can reduce the dimensionality of data without sacrificing important information, mainly focus on extracting image features as the main starting point by finding the low-dimensional space of the data to effectively represent the high-dimensional data features [14], [15]. Traditional DR techniques can be divided into supervised and unsupervised techniques based on the involvement of *a priori* information. Supervised DR techniques are achieved by using *a priori* information to find the optimal low-dimensional subspace. Linear discriminant analysis (LDA) [16] is the most representative algorithm, which maximizes interclass scatter and minimizes intraclass scatter as the primary measure to find the projected subspace. And unsupervised DR techniques are achieved without the involvement of *a priori* information, instead directly learning the features of the data itself. Principal component analysis (PCA) [17] is the classical unsupervised DR technique, which maximizes the variance and finds the optimal orthogonal projection subspace. However, both PCA and LDA are based on the statistical theory and cannot well explore the intrinsic structure of the HSIs [18], while manifold learning methods can effectively mine the manifold structure information of the data. Manifold learning aims to construct a low-dimensional manifold that preserves the geometric relationships between the data [19], [20], [21]. Many manifold learning algorithms have been used for DR in recent years, such as the Laplacian eigenmap (LE) [22], local linear embedding (LLE) [23], local Fisher discriminant analysis (LFDA) [24], local constrained manifold structure collaborative preserving embedding (LMSCPE) [25], and local manifold-based sparse discriminant learning (LMSDL) [20]. LE is based on graph theory and preserves the data neighbor-

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Yao Jin and Yanni Dong are with the Hubei Subsurface Multi-Scale Imaging Key Laboratory, Institute of Geophysics and Geomatics, China University of Geosciences, Wuhan 430074, China, and also with the Key Laboratory of Intelligent Perception and Systems for High-Dimensional Information of Ministry of Education, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: jinyao@cug.edu.cn; dongyanni@cug.edu.cn).

Yuxiang Zhang and Xiangyun Hu are with the Hubei Subsurface Multi-Scale Imaging Key Laboratory, Institute of Geophysics and Geomatics, China University of Geosciences, Wuhan 430074, China (e-mail: zhangyx@cug.edu.cn; xyhu@cug.edu.cn).

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hood structure by using graph embeddings. The goal of LLE is to preserve the original local linear structure of the data in the process of DR. LFDA effectively solves the problem of multimodality and boundary points existing in a class by localizing the intraclass scatter matrix. LMSCPE explores the intrinsic geometric structure in HSIs by combining manifold learning and graph learning in order to uncover the collaborative relationships between sample pairs. LMSDL reveals the manifold structure and complex sparse relationships of HSIs data by mining the manifold-based intersample sparse relationships.

The above DR algorithms have also some limitations; they all ignore the similarity constraints of similar samples and dissimilarity constraints of different classes of samples and do not fully exploit the discriminative information within the data. Metric learning, which can fully exploit the discriminative information of the data [26], aims to obtain an optimal distance matrix that minimizes the distance between similar points and maximizes the distance between dissimilar points. Therefore, the use of metric learning can introduce rich distinguishability information to effectively improve the performance of HSIs in subsequent applications and enhance the discriminative power of the data while reducing its dimensionality. Many metric learning algorithms have been used for DR, such as large margin nearest neighbor (LMNN) [27], information theoretic metric learning (ITML) [28], neighborhood component analysis (NCA) [29], clustered multimetric learning (CMML) [30], and ensemble discriminative local metric learning (EDLML) [31]. LMNN uses a triple constraint to ensure that all points in the neighborhood of a sample point are similar points while keeping the sample point spaced from its dissimilar points. ITML completes the training of the metric matrix by minimizing the relative entropy of the difference between two multivariate Gaussians. The objective of NCA is to maximize the sum of the probabilities of all training samples selecting similar points as their nearest neighbors. CMML combines multitask learning to decompose a single training task into multiple interrelated tasks, enabling effective processing of complex nonlinear data. EDLML performs weighted metric learning for each sample as the nearest neighbor region of the central sample and then integrates all local metrics to ultimately obtain the metric space.

Most metric learning methods directly calculate the Mahalanobis distance between points and then measure the similarity between samples based on this distance. However, this point-to-point distance model ignores the domain information around the sample points and does not consider the structural features of the HSIs, making the Mahalanobis distance vulnerable to the negative effects of noise and extreme points. To solve this problem, our previous work proposed clustered multiple manifold metric learning (CM^3L) [32], which treats a sample point and its immediate neighbors as a manifold, then replaces the point-to-point distance by calculating the distance between the manifolds and improving the discriminative power by considering the structural features of the data. However, the manifold distance in CM^3L only considers the labeled data; it ignores the unlabeled data in HSIs, which destroys the integrity of the constructed manifold. Therefore, how to select suitable neighbors from the unlabeled data to participate in the construction of the manifold is a key issue of this work. Spatial information is important information that HSIs can provide. Many algorithms have been developed to combine spectral and spatial information in order to improve the classification capability of HSIs [33], [34], [35], such as spatial–spectral manifold reconstruction preserving embedding (SSMRPE) [15], discriminant spatial–spectral manifold embedding (DSSME) [36], and coherence-neighborhood preserving embedding (NPE) (SC-NPE) [37]. SSMRPE uses spatial–spectral combined distance to construct a spatial–spectral adjacency graph, which reveals the intrinsic manifold structure of HSIs. DSSME extracts multiple features from the sample points in both the spatial and spectral domains and then linearly combines these features to improve the classification power of HSIs. SC-NPE considers the spatial context of pixels by adopting the difference between the surrounding patches of pixels and then maps the data into the low-dimensional space through an optimized LLE. The above space-spectral combination method only combines the space-spectral information to mine the neighbor relationship of HSIs, ignoring the similarity and dissimilarity constraints between samples. Therefore, it is difficult to mine low-dimensional subspaces with high discriminative ability.

Based on the previous discussion, in order to address the above issues, this study proposes a new spatial–spectral manifold distance (SSMD) for DR and HSIs classification. SSMD combines spectral information with spatial correlation to construct manifolds by selecting neighboring points in labeled and unlabeled samples and then calculating manifold distances to replace point-to-point distances in traditional metric learning.

The main contributions of this article are as follows.

1) In order to make full use of the structural features of HSI and to improve the discriminatory power of the data in the low-dimensional subspace, the SSMD constructs the manifold and calculates the manifold distance by combining the neighboring points with the sample points as the center.
2) Considering the spatial correlation of HSIs, SSMD can select more suitable nearest neighbors for constructing manifolds by combining spatial structure and spectral information to select spatial–spectral neighborhoods, especially for spectrally similar pixels from different classes.
3) To ensure the integrity of the construction of the manifold structure, the SSMD filters the available sample points from both labeled and unlabeled data to participate in the construction of the manifold. In addition, the SSMD sets an adaptive threshold to ensure that the filtered points in the unlabeled data are of the same class as the sample points.

This article is organized as follows. Section II contains some related works, including metric learning and manifold distance. Section III details the proposed SSMD method. Section IV proves the effectiveness of the SSMD through experiments on three real hyperspectral datasets. Finally,

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Section V provides some concluding considerations and a vision for future work.

II. Related Works

A. Metric Learning

In this article, we use $\mathbf{X}_{\text{tr}} = [\mathbf{x}_{\text{tr}_1}, \mathbf{x}_{\text{tr}_2}, \ldots, \mathbf{x}_{\text{tr}_n}] \in R^{d\times n}$ to represent the training sample set, where d is the number of bands and n is the total number of sample points. $\mathbf{Y} = [y_1, y_2, \ldots, y_n] \in R^{1\times n}$ represents the label corresponding to the training sample set. According to the known label information, the training set $\mathbf{X}_{\text{tr}}$ can be divided into two parts. If the classes are the same, then the sample pairs belong to the similar constraint set S; otherwise, the sample pairs belong to the dissimilar constraint set D

$$S : \forall(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_j})\,|_{y_i=y_j} \in S \tag{1}$$

$$D : \forall(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_l})\,|_{y_i\neq y_l} \in D. \tag{2}$$

Metric learning aims to find a metric matrix M that maps the original data into a new feature subspace. The distance between similar points is minimized in the new feature space, and the distance between dissimilar points is maximized [38]. The distance $d_{\mathbf{M}}(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_j})$ between $\mathbf{x}_{\text{tr}_i}$ and $\mathbf{x}_{\text{tr}_j}$ can be computed as

$$d_{\mathbf{M}}(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_j}) = \sqrt{(\mathbf{x}_{\text{tr}_i} - \mathbf{x}_{\text{tr}_j})^{\text{T}}\mathbf{M}(\mathbf{x}_{\text{tr}_i} - \mathbf{x}_{\text{tr}_j})} \tag{3}$$

where $(.)^{\text{T}}$ represents the transpose of the matrix. Then, by constructing similarity constraints and dissimilarity constraints, the metric matrix $\mathbf{M}$ can be obtained

$$\arg\min_{\mathbf{M}} \left(\sum_{\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_j} \in S} d_{\mathbf{M}}\left(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_j}\right) - \sum_{\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_l} \in D} d_{\mathbf{M}}\left(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_l}\right) \right). \tag{4}$$

To solve the high-dimensional number problem, we can find the DR matrix $\mathbf{W} \in R^{d\times m} (m \ll d)$ to map the original high-dimensional data into the low-dimensional space as follows:

$$\mathbf{M} = \mathbf{W}\mathbf{W}^{\text{T}}. \tag{5}$$

Combining (3) and (5), the following equation can be obtained to accomplish both metric learning and DR:

$$\begin{aligned} d_{\mathbf{M}}\left(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_j}\right) &= \sqrt{\left(\mathbf{x}_{\text{tr}_i} - \mathbf{x}_{\text{tr}_j}\right)^{\text{T}}\mathbf{W}\mathbf{W}^{\text{T}}\left(\mathbf{x}_{\text{tr}_i} - \mathbf{x}_{\text{tr}_j}\right)} \\ &= \sqrt{\left(\mathbf{W}^{\text{T}}\mathbf{x}_{\text{tr}_i} - \mathbf{W}^{\text{T}}\mathbf{x}_{\text{tr}_j}\right)^{\text{T}}\left(\mathbf{W}^{\text{T}}\mathbf{x}_{\text{tr}_i} - \mathbf{W}^{\text{T}}\mathbf{x}_{\text{tr}_j}\right)} \\ &= \left\|\mathbf{W}^{\text{T}}\left(\mathbf{x}_{\text{tr}_i} - \mathbf{x}_{\text{tr}_j}\right)\right\|. \end{aligned} \tag{6}$$

B. Manifold Distance

Most metric learning methods directly calculate the Mahalanobis distance between points and then evaluate the similarity between two points by the distance. However, the use of such point-to-point distances ignores the structural features of the data and is also susceptible to the negative effects of noise and extreme points. Therefore, a manifold distance was proposed in CM^{3}L in conjunction with manifold learning. The manifold distance uses the idea of manifold learning to construct a manifold by selecting the nearest suitable neighbors among similar points, with the sample points as the center. Then, the MD is calculated instead of the Mahalanobis distance. MD effectively reduces the effects of noise and extreme data by using the structural information in HSI. The steps of the MD calculation are as follows. First, for each training sample $\mathbf{x}_{\text{tr}_i}$, the nearest k points are selected by the Euclidean distance. Then, we use the training sample as the centroid and combine its nearest neighbors to construct the manifold $S_i = [\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_{i1}}, \ldots, \mathbf{x}_{\text{tr}_{ik}}] \in R^{d\times(k+1)}$. Finally, the distance between the manifolds is calculated as follows:

$$\begin{aligned} &d^2_{\text{manifold}^{(c)}}\left(S_i, S_j\right) \\ &\quad = d^2_{\mathbf{M}^{(c)}}\left(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_j}\right) \\ &\qquad + \theta \sum_{q=1}^{k}\left(d^2_{\mathbf{M}^{(c)}}\left(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_{jq}}\right) + d^2_{\mathbf{M}^{(c)}}\left(\mathbf{x}_{\text{tr}_j}, \mathbf{x}_{\text{tr}_{iq}}\right)\right). \end{aligned} \tag{7}$$

The first term in the formula calculates the distance between the central sample points of the manifold. The second term calculates the distance between the nearest neighbor point in the manifold and the center point of the other manifold. Furthermore, we set the hyperparameter θ to balance the influence of sample points and nearest neighbor points on the SSMD.

In addition, considering that different nearest neighbors make different contributions to the construction of the manifold, we use the reconstruction coefficient to measure the similarity between the sample and the nearest neighbors, and avoid ignoring those nearest neighbors that provide more information. This is calculated as follows:

$$\arg\min_{w} \sum_{i=1}^{N} \left\| \mathbf{x}_i - \sum_{q=1}^{k} w_{iq}\mathbf{x}_{\text{tr}_{iq}} \right\|. \tag{8}$$

The reconstruction coefficient w_{iq} measures the degree of similarity between a sample point and its neighbors, where if the degree of similarity is higher, then it means that the nearest neighbor provides more information and is, therefore, assigned a higher weight (where the converse is also true). Thus, (7) can be rewritten as

$$\begin{aligned} &d^2_{\text{manifold}^{(c)}}\left(S_i, S_j\right) \\ &\quad = d^2_{\mathbf{M}^{(c)}}\left(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_j}\right) \\ &\qquad + \theta \sum_{q=1}^{k}\left(w_{jq}d^2_{\mathbf{M}^{(c)}}\left(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_{jq}}\right) + w_{iq}d^2_{\mathbf{M}^{(c)}}\left(\mathbf{x}_{\text{tr}_j}, \mathbf{x}_{\text{tr}_{iq}}\right)\right). \end{aligned} \tag{9}$$

III. Proposed Method

Although the MD in CM^{3}L considers the structural features of the HSIs, it ignores the information in the unlabeled data, resulting in an incomplete manifold structure. Therefore, we propose SSMD based on the manifold distance, which constructs a manifold by using spatial–spectral information to screen suitable neighbors in both labeled and unlabeled data. The flowchart for the SSMD calculation is shown in Fig. 1. First, the nearest neighbor point is selected among the labeled and unlabeled samples based on spatial–spectral information

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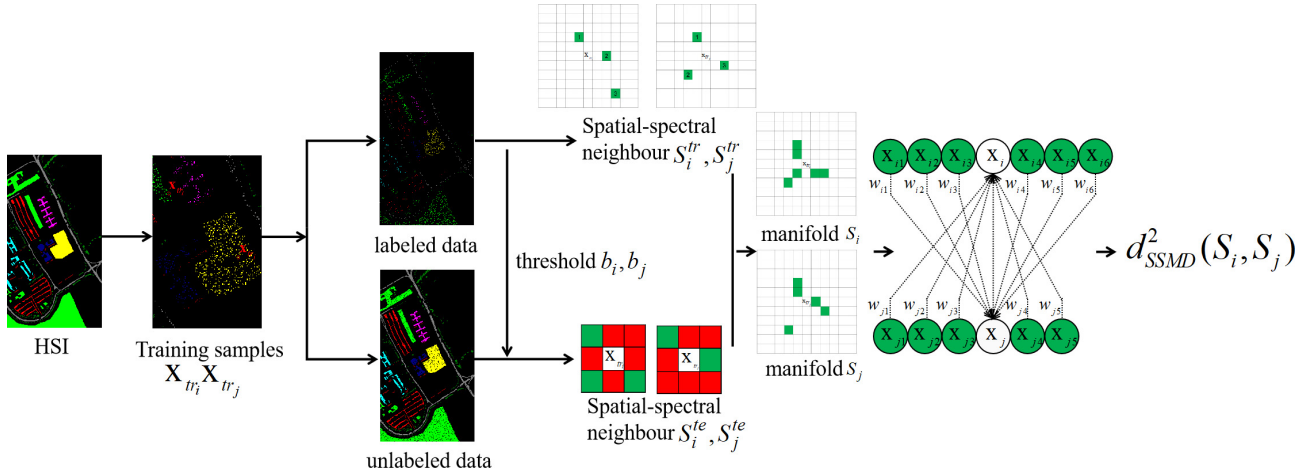


Fig. 1. Flowchart of SSMD. First, suitable nearest neighbors are selected from the labeled and unlabeled data based on the spatial–spectral information to participate in the construction of the manifold. Then, the weighted distances between the center points and the nearest neighbors of the manifold are calculated separately to obtain the SSMD.

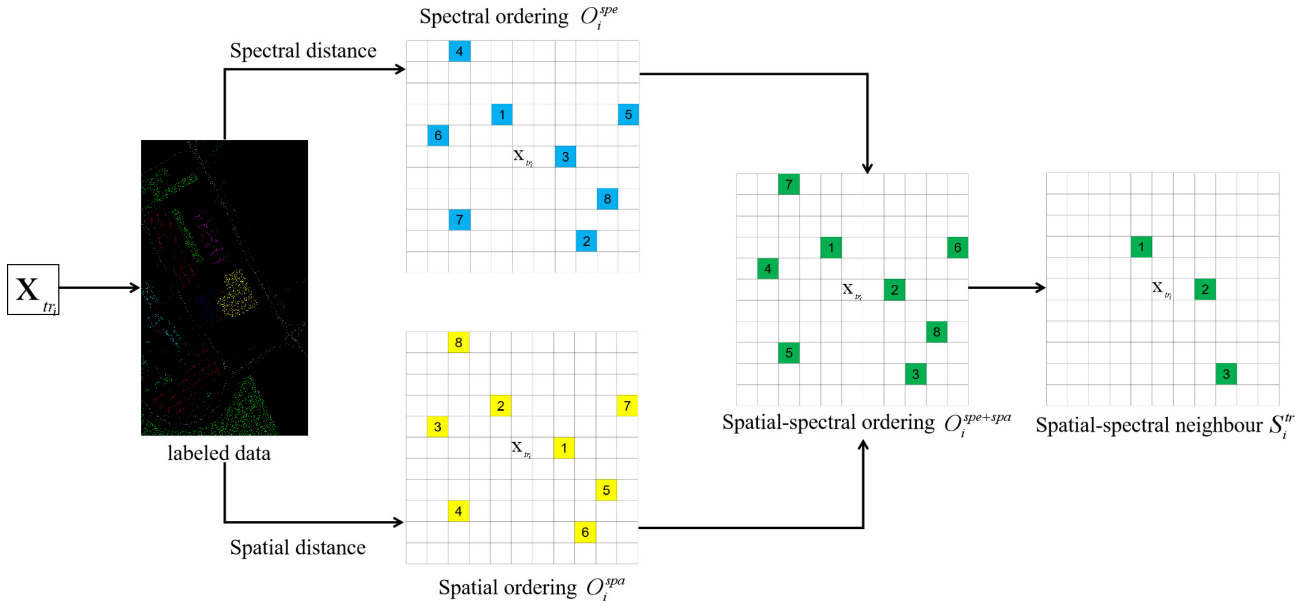


Fig. 2. Flowchart for selecting spatial–spectral nearest neighbors in labeled data. First calculate the spatial distances and spectral distances between sample points and similar points to obtain the spatial ordering and spectral ordering, respectively, and then integrate the spatial ordering and spectral ordering to obtain the spatial–spectral ordering, where the k_1 points with the smallest ordering are the spatial–spectral nearest neighbor points.

with the sample point as the center. Then, the central sample point and the screened nearest neighbor points are considered as a whole in order to construct the manifold. Finally, the weighted distances between the central and neighbor points of the manifold are calculated separately to obtain the SSMD.

A. Selection of Spatial–Spectral Neighbors

According to the first law of geology, there is a spatial correlation between pixels, so the spatial distribution of pixels and features is consistent within a certain range. Therefore, by fusing the spatial and spectral information in HSIs, the similarity between pixels can be effectively measured, and more suitable nearest neighbors can be selected. In this article, in order to construct more suitable manifolds, SSMD uses spatial–spectral information as a criterion to screen for suitable nearest neighbors in both labeled and unlabeled data to participate in the construction of manifolds.

The sample points for constructing the manifold in SSMD consist of two parts except for the central sample point, one is the nearest neighbors filtered from the labeled data and the other is the nearest neighbors filtered from the unlabeled data. Both parts are selected based on the spatial–spectral information, and different selection methods are used because the two parts of the data contain different *a priori* information.

1) Selection of Spatial–Spectral Neighbors in Labeled Data: First, the nearest neighbor points are selected in the labeled data since the labeled data contain the labeling information, so the selection can be made directly among similar sample points based on the spectral–spatial information. The specific process for selecting spatial–spectral nearest neighbors in the labeled data is shown in Fig. 2. For each training sample $\mathbf{x}_{\mathrm{tr}_i}$,

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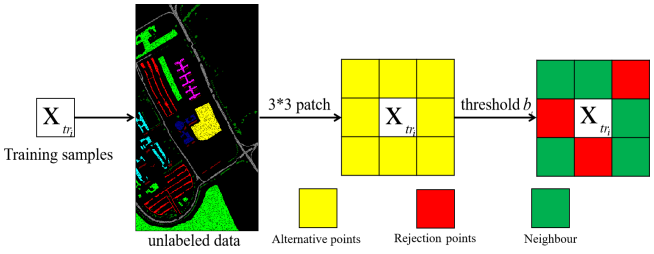


Fig. 3. Flowchart for selecting spatial–spectral nearest neighbors in unlabeled data. First, select alternative points in the unlabeled data according to the 3 × 3 patches, and then eliminate the points larger than the adaptive threshold b_i. The remaining points are the spatial–spectral nearest neighbor points.

the spectral and spatial distances between the sample points and other similar sample points were calculated separately by using the Euclidean distances. The spectral distances are first calculated by using the following formula:

$$d_{\text{spe}}(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_j}) = \left\|\mathbf{x}_{\text{tr}_i} - \mathbf{x}_{\text{tr}_j}\right\|_2 \tag{10}$$

and then, the ordinal number $O_i^{\text{spe}} = [o_1^{\text{spe}}, o_2^{\text{spe}}, \ldots, o_n^{\text{spe}}]$ corresponding to each sample point is calculated by sorting the spectral distances from smallest to largest.

Denote the coordinates of $\mathbf{x}_{\text{tr}_i}$ and $\mathbf{x}_{\text{tr}_j}$ in the HSIs as (p_i, q_i) and (p_j, q_j); then, the spatial coordinate distance $d_{\text{spa}}(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_j})$ is given by the Euclidean distance between their coordinates as

$$d_{\text{spa}}(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_j}) = \sqrt{(p_i - p_j)^2 + (q_i - q_j)^2}. \tag{11}$$

Similarly, a new ordinal number $O_i^{\text{spa}} = [o_1^{\text{spa}}, o_2^{\text{spa}}, \ldots, o_n^{\text{spa}}]$ can be obtained based on the spectral distance from smallest to largest. Then, the two ordinations O_i^{spe}, O_i^{spa} can be summed to obtain an ordination $O_i^{\text{spe+spa}} = [o_1^{\text{spe}}+o_1^{\text{spa}}, o_1^{\text{spe}}+o_2^{\text{spa}}, \ldots, o_1^{\text{spe}}+o_n^{\text{spa}}]$ based on spatial–spectral information. After sorting their ordinations from smallest to largest, the sample points corresponding to the smallest k_1 ordinations are taken as the selected spatial–spectral nearest neighbors $S_i^{\text{tr}} = [\mathbf{x}_{\text{tr}_{i1}}, \ldots, \mathbf{x}_{\text{tr}_{ik_1}}] \in R^{d\times k_1}$.

2) Selection of Spatial–Spectral Neighbors in Unlabeled Data: When selecting spatial–spectral nearest neighbors in unlabeled data, alternative points are first selected using a 3 × 3 patch, with sample points at the center of the patch and the other samples in the patch belonging to the alternative points. In this way, for each training sample, up to eight alternative points can be selected based on the spatial information, and if the sample points are at the edges of the image, then the number of alternative points is less than 8. The specific process for selecting spatial–spectral nearest neighbors in the unlabeled data is shown in Fig. 3. Since the unlabeled data lack labeled information, it is not possible to determine the class of alternative points or to filter out similar points to participate in the construction of the manifold, so a threshold b is introduced to constrain the selection of the final nearest neighbor points. If $d_{\text{spe}}(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_j}) > b$, then $\mathbf{x}_{\text{tr}_i}$ and $\mathbf{x}_{\text{tr}_j}$ belong to different classes, and if $d_{\text{spe}}(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_j}) < b$, then $\mathbf{x}_{\text{tr}_i}$ and $\mathbf{x}_{\text{tr}_j}$ belong to the same class.

In the training process, each sample point contains different neighborhood information. Therefore, in the face of complex data, setting a fixed threshold does not consider the variability among sample points and often fails to achieve good classification results. In this article, we use an adaptive threshold to filter the alternative points. The adaptive threshold is set according to the neighborhood information of the sample points, which fully considers the variability among the sample points and achieves a specific analysis for specific problems. The method sets different thresholds for the sample points, specifically considering the distance between a sample point and its similar points in the labeled data. In this article, we set the threshold $b_i = \min d_{\text{spe}}(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_{ik}})$, where $\mathbf{x}_{\text{tr}_{ik}} \in S_i^{\text{tr}}$. Then, the distance between the sample point and the alternative points is calculated, and alternative points whose distances are within the threshold b_i are used as the neighbor points. Finally, for each sample point $\mathbf{x}_{\text{tr}_i}$, we can obtain k_2 spatial–spectral neighbors $S_i^{\text{te}} = [\mathbf{x}_{\text{te}_{i1}}, \ldots, \mathbf{x}_{\text{te}_{ik_2}}] \in R^{d\times k_2}$ in the unlabeled data. Unlike S_i^{tr}, here the value of k_2 is not fixed, and each sample point corresponds to a different value of k_2. The value of k_2 depends on the number of points among the alternatives that satisfy the threshold constraint.

B. Weighted Manifold Distance Calculation

Based on the spatial–spectral information, we selected k_1 and k_2 neighbors from the labeled and unlabeled data, respectively. The selected neighboring points S_i^{tr} and S_i^{te} are integrated to jointly participate in the construction of manifold $S_i = [\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_{i1}}, \ldots, \mathbf{x}_{\text{tr}_{ik_1}}, \mathbf{x}_{\text{te}_{i1}}, \ldots, \mathbf{x}_{\text{te}_{ik_2}}] \in R^{d\times(1+k_1+k_2)}$.

Similar to the manifold distance in CM^{3}L, we consider different contributions made by different neighboring points in the process of constructing the manifold. In this article, the reconstruction coefficient w_{iq} is used to measure the similarity between the nearest neighbor and the sample points. If the similarity is higher, then the reconstruction coefficient is also higher, and vice versa. The reconstruction coefficient is calculated using (8). The specific calculation formula of the SSMD is shown in (12). After calculating the SSMD, it is used to replace the Mahalanobis distance, and an improved metric learning algorithm can be obtained. Algorithm 1 illustrates the steps of the proposed SSMD

$$\begin{aligned} &d^2_{\text{SSMD}}(S_i, S_j) \\ &= d^2_{\mathbf{M}}(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_j}) \\ &\quad + \theta\left(\sum_{q=1}^{k_1}\left(w_{jq}d^2_{\mathbf{M}}(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{tr}_{jq}}) + w_{iq}d^2_{\mathbf{M}}(\mathbf{x}_{\text{tr}_j}, \mathbf{x}_{\text{tr}_{iq}})\right)\right. \\ &\quad \left. + \sum_{q=1}^{k_2}\left(w_{j(k_1+q)}d^2_{\mathbf{M}}(\mathbf{x}_{\text{tr}_i}, \mathbf{x}_{\text{te}_{jq}}) + w_{i(k_1+q)}d^2_{\mathbf{M}}(\mathbf{x}_{\text{tr}_j}, \mathbf{x}_{\text{te}_{iq}})\right)\right). \end{aligned} \tag{12}$$

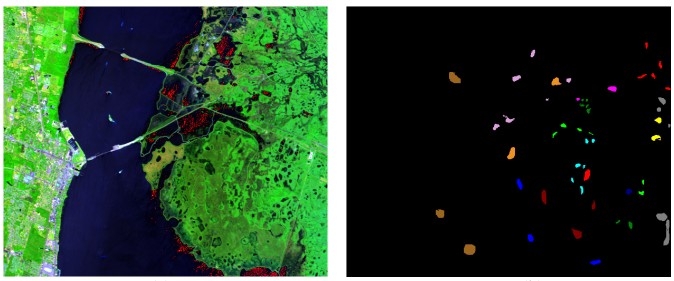
IV. Experiments

In this section, seven groups of comparison experiments are set up and are compared with the original algorithm after replacing the traditional Mahalanobis distance with the SSMD

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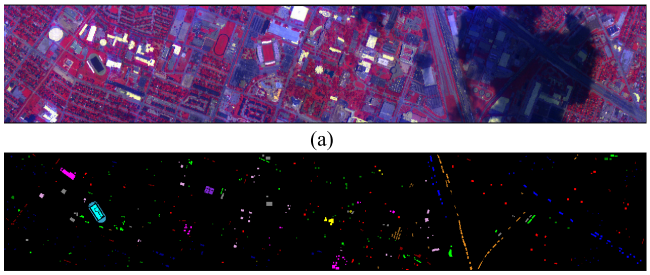
Algorithm 1 Procedure of the Proposed SSMD

Input: Training samples: $X_{tr} = [\mathbf{x}_{tr_1}, \mathbf{x}_{tr_2}, \ldots, \mathbf{x}_{tr_{n1}}] \in \mathrm{R}^{d \times n_1}$
Test samples: $X_{te} = [\mathbf{x}_{te_1}, \mathbf{x}_{te_2}, \ldots, \mathbf{x}_{te_{n2}}] \in \mathrm{R}^{d \times n_2}$
Training labels: $Y = [y_1, y_2, \ldots, y_{n_1}] \in \mathrm{R}^{1 \times n_1}$
Parameters: k_1, θ
Steps: For each training sample $\mathbf{x}_{tr_i}$, do
1) Select k_1 neighborhoods for $\mathbf{x}_{tr_i}$ in labeled data
2) Select k_2 neighborhoods for $\mathbf{x}_{tr_i}$ in unlabeled data
3) Construct the manifold S_i for $\mathbf{x}_{tr_i}$
4) Calculate the coefficient w_{iq} according to Formula (8).
5) Calculate $d^2_{SSMD}(S_i, S_j)$ according to Formula (12).
Output: Spatial-spectral manifold distance $d^2_{SSMD}(S_i, S_j)$



(a) (b)

Fig. 4. (a) False-color image of the KSC dataset. (b) Ground-truth map.



(a) (b)

Fig. 5. (a) False-color image of the Houston dataset. (b) Ground-truth map.

based on seven metric learning algorithms. In addition, in the seventh set of comparison experiments, we add a comparison with CM^{3}L as a way to demonstrate the superiority of SSMD over MD. Experiments have been conducted on three real HSI datasets to validate the effectiveness of the proposed SSMD.

A. Dataset Description

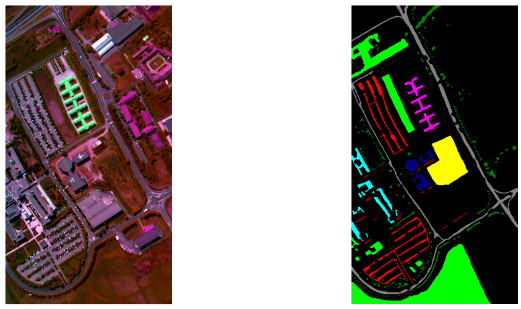
The first dataset is the KSC dataset [39]. The KSC dataset was collected by NASA's Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) instrument at the Kennedy Space Center (KSC), Merritt Island, FL, USA, in 1996. The dataset includes 224 spectral bands in the wavelength range from 0.4 to 2.5 μm, leaving a total of 176 bands after removing the low signal-to-noise and water absorption bands. The corresponding false-color image ground-truth map and ground-truth map are shown in Fig. 4(a) and (b), respectively. The image size of the KSC dataset is 512 $\times$ 614, and there are 5211 sample points in 13 classes of features. In this article, 528 sample points were randomly selected as training samples in this experiment, and the remaining 4683 sample points were used as the test samples. The number of samples of each class used in the experiment for all algorithms is shown in Table I.

TABLE I
NUMBER OF SAMPLES OF THE KSC DATASET USED FOR EACH CLASS

No.	Class	Training	Test	No.	Class	Training	Test
C1	Scrub	77	684	C8	Graminoid marsh	44	387
C2	Willow swamp	25	218	C9	Spartina marsh	52	468
C3	CP hammock	26	230	C10	Catiail marsh	41	363
C4	CP/Oak	26	226	C11	Salt marsh	42	377
C5	Slash pine	17	144	C12	Mud flats	51	452
C6	Oak/Broad leaf	23	206	C13	Water	93	834
C7	Hard-wood swamp	11	94				
					Total	**528**	**4683**

TABLE II
NUMBER OF SAMPLES OF THE HOUSTON DATASET USED FOR EACH CLASS

No.	Class	Training	Test	No.	Class	Training	Test
C1	Healthy grass	126	1125	C9	Road	126	1129
C2	Stressed grass	126	1128	C10	Highway	123	1104
C3	Synthetic grass	70	627	C11	Railway	124	1111
C4	Trees	125	1119	C12	Parking Lot 1	124	1109
C5	Soil	125	1117	C13	Parking Lot 2	47	422
C6	Water	33	292	C14	Tennis Court	43	385
C7	Residential	127	1141	C15	Running Track	66	594
C8	Commercial	125	1119				
					Total	**1510**	**13519**



(a) (b)

Fig. 6. (a) False-color image of the Pavia University dataset. (b) Ground-truth map.

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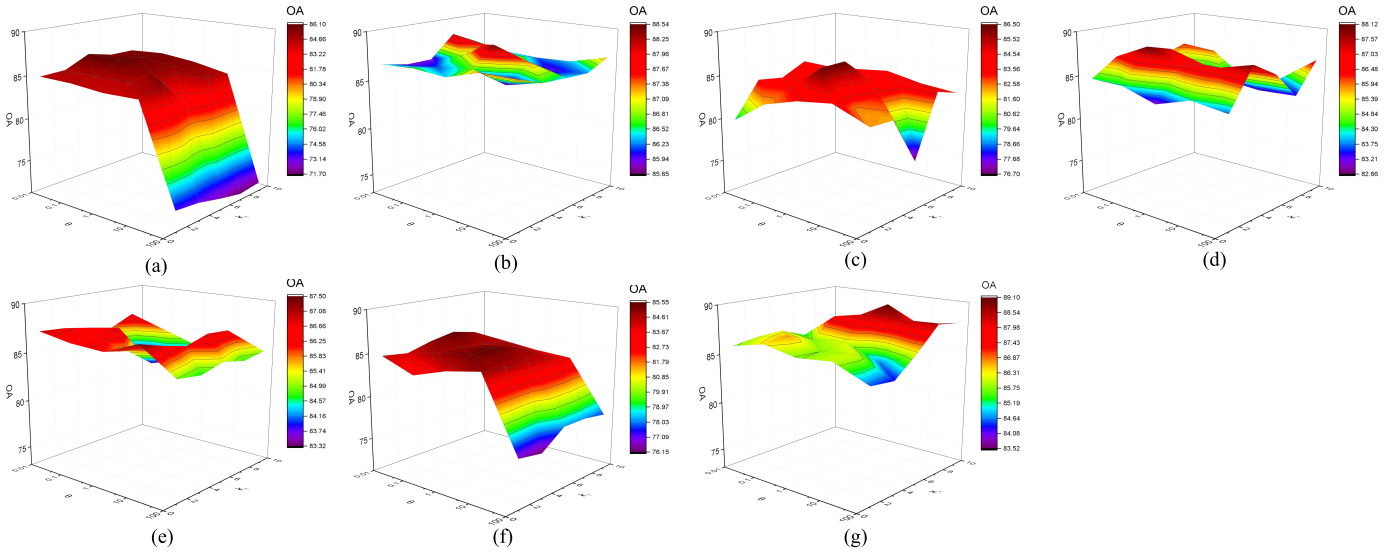


Fig. 7. Overall accuracy (OA) versus different parameters k_1 and θ with the KSC dataset: (a) LMNN; (b) ITML; (c) MCML; (d) NCA; (e) AWLML; (f) ML-CC; and (g) CMML.

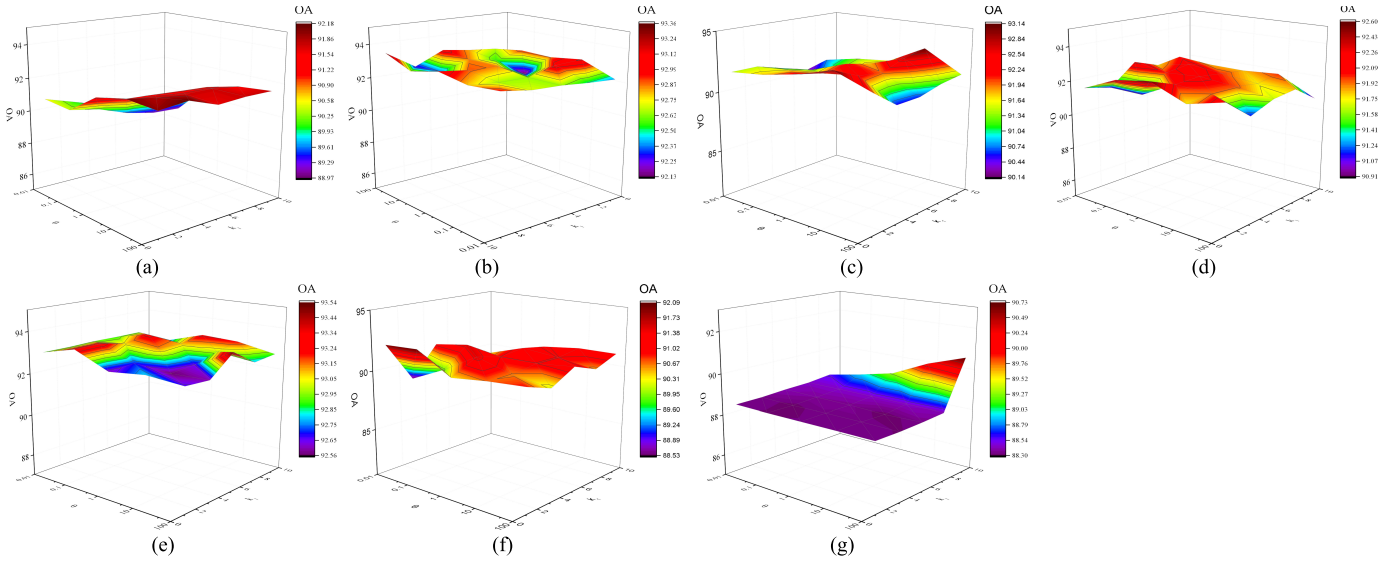


Fig. 8. OA versus different parameters k_1 and θ with the Houston dataset: (a) LMNN; (b) ITML; (c) MCML; (d) NCA; (e) AWLML; (f) ML-CC; and (g) CMML.

The second dataset is the Houston dataset [40]. The Houston dataset is the University of Houston campus and adjacent urban areas captured by the National Science Foundation-funded Center for Airborne Laser Mapping (NCALM). The HSI has 144 spectral bands in the spectral range of 380–1050 nm. The image size is 349 × 1905, and a total of 15 classes of 15 029 pixels were selected for the experiment in this article. A total of 1510 pixels out of the 15 029 pixels were considered to be training samples, and the remaining pixels were used as the test samples. The specific number of samples for each class is shown in Table II. Fig. 5(a) and (b) shows the false-color image ground-truth map and ground-truth map, respectively.

The third dataset is the Pavia University dataset [41]. This HSI dataset was collected by the Reflective Optics System Imaging Spectrometer (ROSIS) over the Italian city of Pavia in 2003. The spatial resolution of the image is 1.3 m, with a total of 115 bands, and after removing the 12 bands that had noise effects, the remaining 103 bands are used for the experiments in this article. The size of the data is 610 × 340 pixels, so it contains a total of 2 207 400 pixels for nine types of features. The number of samples of each class used in the experiment for all algorithms is shown in Table III. Fig. 6(a) and (b) shows the false-color image ground-truth map and ground-truth map, respectively.

B. Experimental Setting

In this section, we provide some explanations of the experimental setup. To verify the DR effect of the algorithm,

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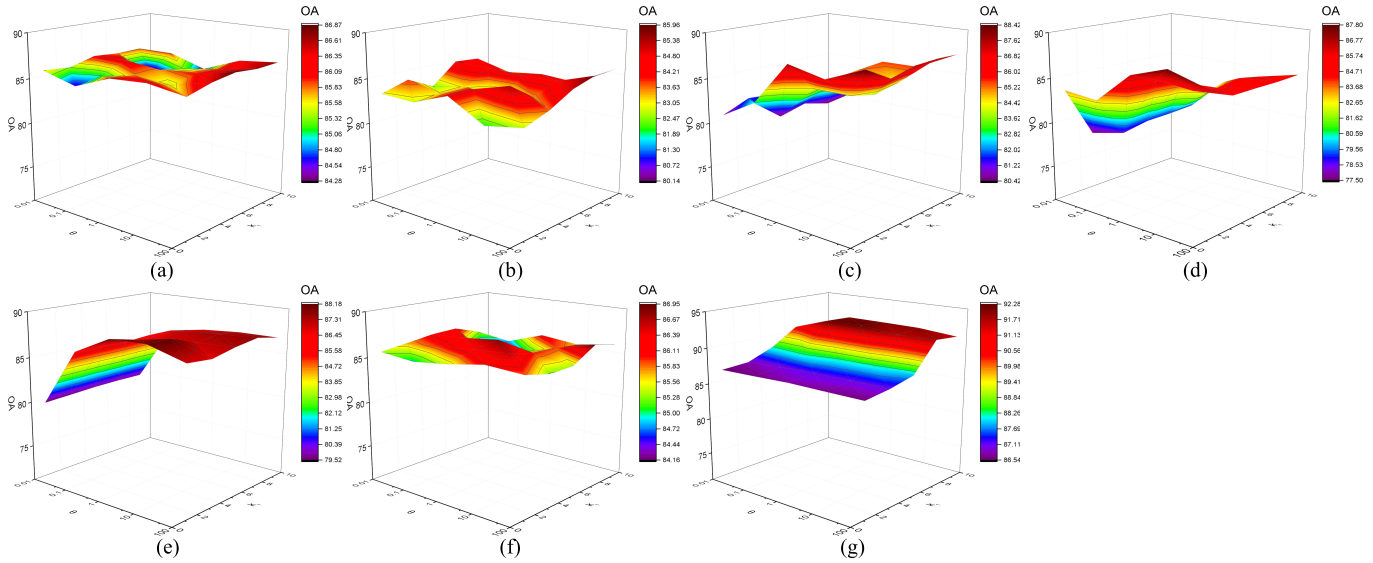


Fig. 9. OA versus different parameters k_1 and θ with the Pavia University dataset: (a) LMNN; (b) ITML; (c) MCML; (d) NCA; (e) AWLML; (f) ML-CC; and (g) CMML.

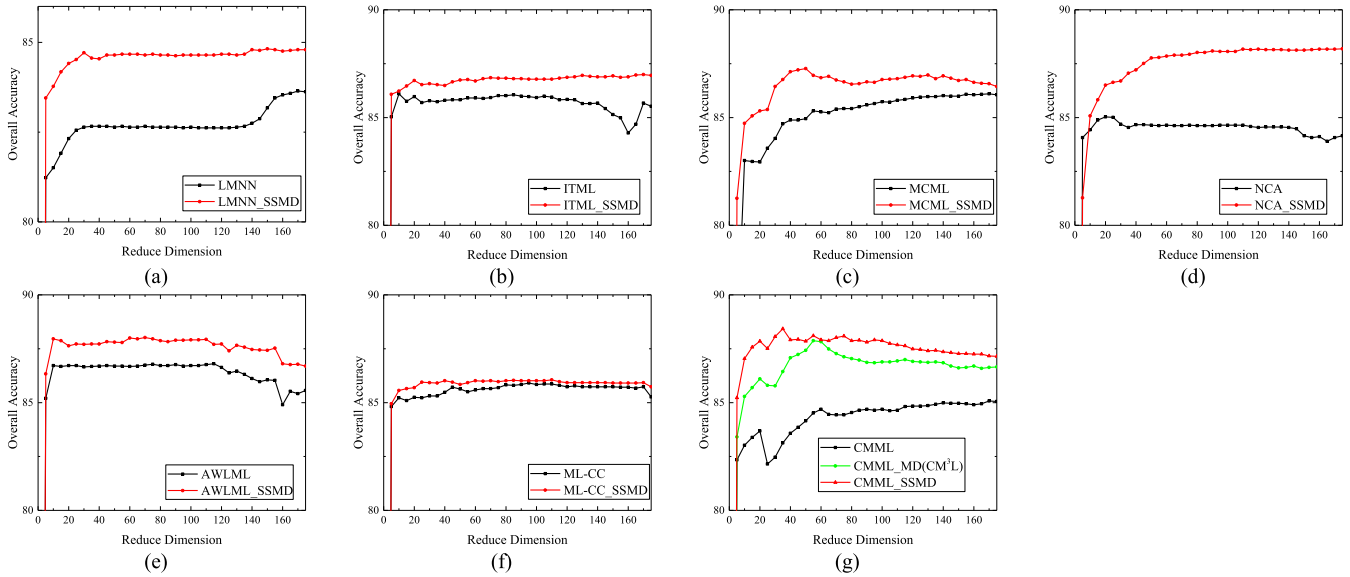


Fig. 10. Classification OA with respect to reduced dimensionality with the KSC dataset: (a) LMNN; (b) ITML; (c) MCML; (d) NCA; (e) AWLML; (f) ML-CC; and (g) CMML.

TABLE III

NUMBER OF SAMPLES OF THE PAVIA UNIVERSITY DATASET USED FOR EACH CLASS

No.	Class	Training	Test	No.	Class	Training	Test
C1	Asphalt	664	5967	C6	Bare Soil	503	4526
C2	Meadows	1865	16784	C7	Bitumen	133	1197
C3	Gravel	210	1889	C8	Bricks	369	3313
C4	Trees	307	2757	C9	Shadows	95	852
C5	Metal sheets	135	1210				
					Total	**4281**	**38459**

we use the 1-nearest neighbor (1NN) classifier to classify the data after DR and then measure the effectiveness of the DR algorithm by classification accuracy. First, we chose seven metric learning algorithms as the base algorithms, including CMML, LMNN, ITML, maximally collapsing metric learning (MCML), NCA, asymmetric weighted logistic metric learning (AWLML) [42], and the metric learning model with clustering-based constraints (ML-CC) [43]. The original seven metric learning algorithms are compared with the seven corresponding new metric learning algorithms obtained after using SSMD to verify their effectiveness of SSMD. In addition, since CM^{3}L was obtained by adding MD to CMML, we added a comparison of the CMML_SSMD algorithm with CM^{3}L in the seventh set of comparison experiments to demonstrate the superiority of SSMD over MD.

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TABLE IV
CLASSIFICATION ACCURACIES FOR THE KSC DATASET (DIM = 40)

Class	LMNN		ITML		MCML		NCA		AWLML		ML-CC		CMML		
	-	SSMD	-	SSMD	-	SSMD	-	SSMD	-	SSMD	-	SSMD	-	MD	SSMD
C1	87.43 ±0.22	**89.62 ±0.24**	90.79 ±0.53	**91.37 ±0.28**	**88.16 ±0.19**	86.11 ±0.59	83.92 ±0.43	**86.11 ±0.59**	88.16 ±0.52	**90.64 ±0.57**	**84.80 ±0.66**	84.50 ±0.51	89.33 ±0.70	**91.67 ±0.54**	88.45 ±0.42
C2	69.72 ±0.41	**70.18 ±0.61**	**88.99 ±0.50**	87.61 ±0.41	77.52 ±1.63	**88.07 ±1.35**	82.11 ±0.41	**84.86 ±0.20**	**88.99 ±0.95**	87.61 ±0.69	**87.61 ±0.99**	85.78 ±1.00	76.61 ±0.32	82.57 ±1.02	**85.32 ±2.08**
C3	86.09 ±2.19	**87.83 ±2.31**	74.35 ±4.58	**86.09 ±4.45**	81.74 ±1.70	**91.74 ±2.31**	82.17 ±0.19	**82.17 ±0.19**	**81.74 ±2.00**	80.00 ±2.28	82.61 ±2.55	**82.61 ±2.63**	88.70 ±3.03	88.26 ±2.00	**92.61 ±2.05**
C4	43.36 ±1.11	**44.25 ±0.06**	53.45 ±0.91	**53.54 ±4.31**	55.75 ±3.59	**64.60 ±1.53**	**60.18 ±0.59**	53.10 ±3.13	63.27 ±0.95	**63.72 ±2.13**	54.42 ±3.02	**60.62 ±2.45**	56.64 ±2.04	56.19 ±1.75	**63.27 ±3.00**
C5	40.97 ±0.38	**42.36 ±1.05**	51.39 ±1.05	**55.56 ±2.62**	42.36 ±1.29	**45.83 ±3.45**	65.97 ±0.62	**69.44 ±2.92**	45.83 ±2.16	**46.53 ±0.90**	63.19 ±0.62	**64.58 ±2.70**	43.06 ±5.07	**61.11 ±1.87**	59.03 ±0.90
C6	40.29 ±	**41.26 ±0.43**	**44.17 ±3.02**	39.32 ±2.67	45.15 ±0.86	**52.91 ±1.89**	38.35 ±1.00	**56.80 ±2.28**	48.54 ±3.90	**51.94 ±2.89**	39.32 ±2.67	**39.81 ±1.41**	41.26 ±2.48	**50.49 ±3.70**	50.00 ±1.00
C7	93.62 ±0.58	**94.68 ±0.11**	84.04 ±4.18	**86.17 ±1.90**	**86.17 ±1.93**	78.72 ±5.02	81.91 ±0.95	**89.36 ±2.53**	76.60 ±2.85	**77.66 ±4.07**	87.23 ±0.58	**88.30 ±0.47**	76.60 ±2.33	**76.60 ±2.64**	73.40 ±2.75
C8	82.17 ±0.14	**87.34 ±0.61**	78.29 ±1.24	**84.50 0.72**	84.24 ±1.16	**87.60 ±2.50**	82.43 ±1.53	**83.20 ±0.56**	80.62 ±0.59	**82.43 ±0.95**	83.98 ±1.46	**85.27 ±1.17**	79.59 ±0.82	83.20 ±0.59	**85.53 ±0.83**
C9	88.89 ±0.21	**91.24 ±0.17**	94.02 ±0.57	**94.87 ±0.41**	**95.09 ±0.23**	94.66 ±0.80	96.15 ±0.05	**98.08 ±0.24**	98.08 ±0.95	**98.29 ±0.57**	94.66 ±0.23	**95.09 ±0.17**	91.03 ±0.38	94.44 ±0.17	**96.58 ±1.31**
C10	77.96 ±0.33	**86.50 ±0.23**	**89.81 ±0.23**	89.26 ±1.44	83.75 ±0.73	**92.29 ±0.64**	87.60 ±0.36	**91.46 ±0.38**	91.18 ±0.90	**93.39 ±0.57**	85.67 ±0.69	**86.50 ±1.01**	84.30 ±4.09	90.63 ±0.81	**95.04 ±1.55**
C11	93.63 ±0.14	**93.63 ±0.47**	89.12 ±0.45	**95.49 ±0.68**	**96.55 ±0.23**	94.96 ±0.58	94.16 ±0.23	**97.08 ±0.29**	**96.02 ±0.55**	94.96 ±0.29	**96.55 ±0.14**	96.29 ±0.40	91.95 ±0.43	**93.37 ±0.47**	92.31 ±0.26
C12	84.96	**86.95**	**92.70**	90.93	83.41	**85.18**	87.39	**91.59**	86.28	**92.04**	90.71	**90.71**	83.63	87.61	**90.27**
	±0.19	**±0.47**	**±1.12**	±0.59	±0.98	**±1.55**	±0.47	**±2.01**	±1.82	**±1.36**	±0.34	**±0.33**	±0.82	±1.23	**±0.69**
C13	98.92 ±0.00	**98.92 ±0.00**	98.92 ±0.10	**99.04 ±0.06**	**99.64 ±0.05**	99.52 ±0.19	97.96 ±0.00	**98.44 ±0.06**	98.92 ±0.09	**99.04 ±0.00**	98.08 ±0.05	**98.08 ±0.03**	99.04 ±0.19	98.92 ±0.06	**98.92 ±0.19**
OA	82.31 ±0.25	**84.41 ±0.22**	85.41 ±0.21	**86.88 ±0.33**	84.73 ±0.41	**87.05 ±0.36**	85.05 ±0.16	**87.50 ±0.70**	86.37 ±0.59	**87.63 ±0.25**	85.60 ±0.41	**86.05 ±0.46**	83.66 ±0.48	86.84 ±0.51	**87.82 ±0.75**
kappa	0.8030 ± 0.0030	**0.8263 ± 0.0025**	0.8375 ± 0.0024	**0.8540 ± 0.0037**	0.8300 ± 0.0060	**0.8560 ± 0.0041**	0.8339 ± 0.0018	**0.8610 ± 0.0079**	0.8484 ± 0.0066	**0.8624 ± 0.0028**	0.8400 ± 0.0047	**0.8450 ± 0.0051**	0.8182 ± 0.0055	0.8535 ± 0.0058	**0.8646 ± 0.0084**

In this article, we randomly select 10% of each class in each image as the training samples, while the rest are used as the test samples for experimental performance validation to ensure the fairness of comparison. In the meantime, each classification is performed five times to obtain the average classification accuracy and reduce the possible deviation caused by random sampling.

In addition, to ensure a fair comparison between the improved algorithm and the original algorithm, the corresponding algorithm adopts the same parameter settings. For the proposed SSMD, there are two parameters, including the number of neighboring points k_1 in the labeled data and a hyperparameter θ. From Figs. 7–9, we can determine that the classification results are more stable for different values of k_1, and there are good classification results using SSMD when $\theta = 10$, so we set $k_1 = 3$, $\theta = 10$.

C. Performance on the Three Datasets

1) KSC Dataset: Fig. 10 represents a figure of the overall classification accuracy for different feature subspace dimensions obtained using the 1NN classifier. The black lines in Fig. 10 represent the results of the original algorithm, and the red lines represent the results obtained by the original algorithm after using the SSMD. From Fig. 10, it can be determined that the results of the algorithms were all improved, to a certain extent, after using SSMD, with CMML, MCML, and NCA improving by greater amounts, while AWLML and ML-CC improved relatively less. Similarly, the classification ability of the algorithm is improved by using SSMD to replace MD in the seventh group of comparison experiments.

From Table IV, we can see that all new algorithms using SSMD achieved good results in the seven comparison groups, with OA values increasing by up to 3%. In the seventh group of comparison experiments, we found that the new algorithm with SSMD has an improved accuracy compared to the algorithm with MD, which also proves that SSMD is better than MD.

In this article, in order to show the classification results more directly, Fig. 11 shows a map of the classification results of all algorithms for the KSC dataset. From Fig. 11, we can see that the new algorithms have better classification results in all seven comparison groups, especially class 5 (Slash pine), class 6 (Oak/Broad leaf), and class 7 (Hard-wood swamp).

2) Houston Dataset: Fig. 12 shows the classification OA values of all methods based on the 1NN classifier in different dimensions. We can see from Fig. 12 that the new algorithm with SSMD showed a large improvement in terms of the results for all seven groups of comparison experiments. All the new algorithms have better classification

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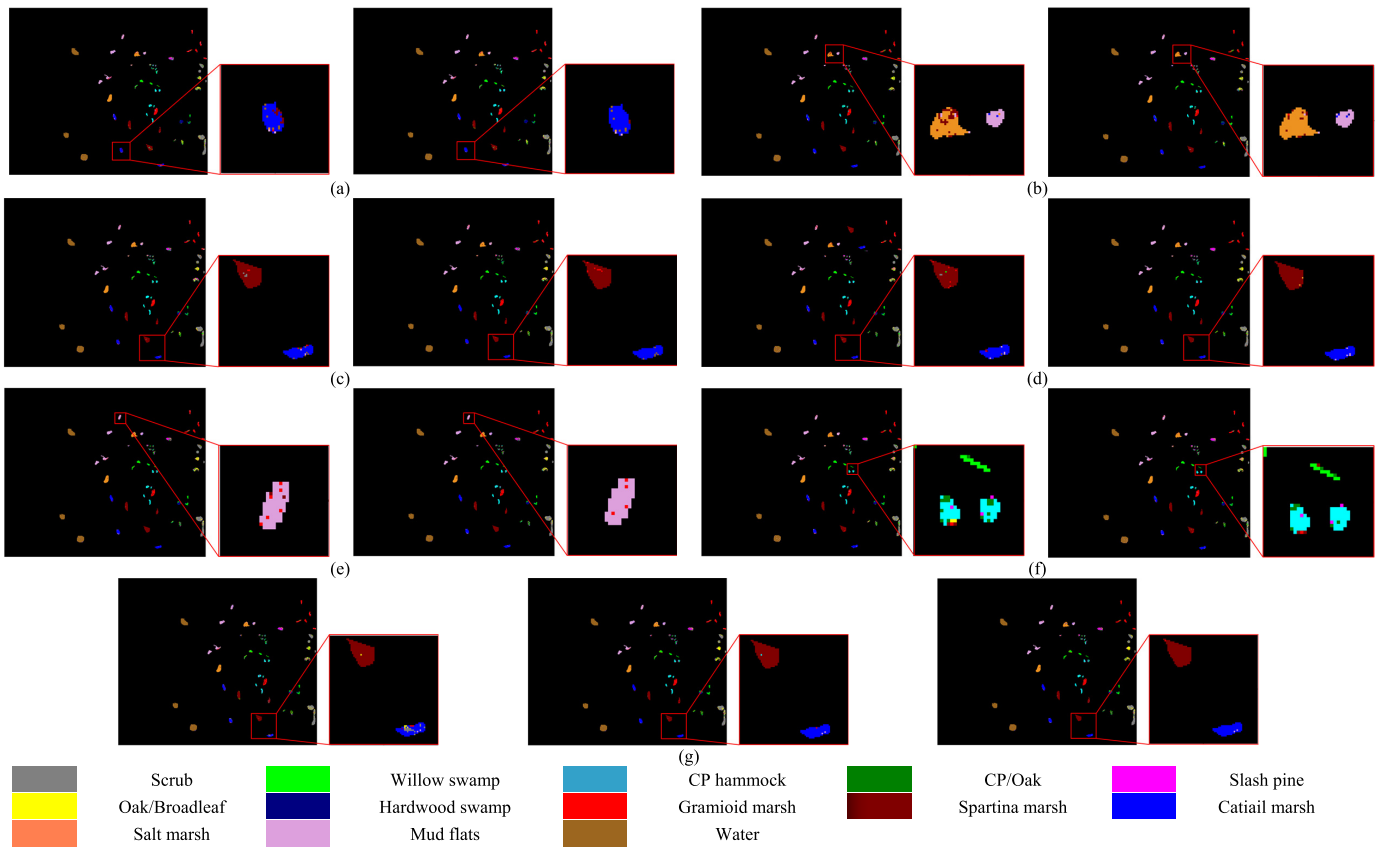


Fig. 11. Classification results for all the methods with the KSC dataset based on 1NN: (a) LMNN; (b) ITML; (c) MCML; (d) NCA; (e) AWLML; (f) ML-CC; and (g) CMML (Left: original and Right: new).

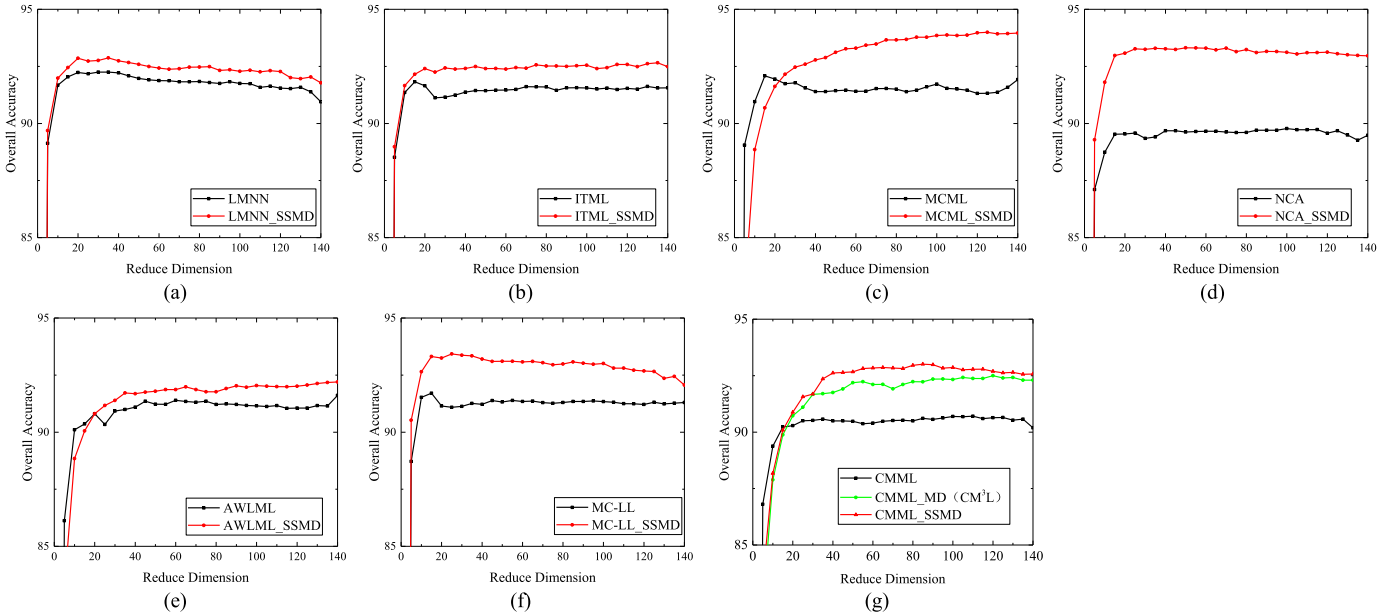


Fig. 12. Classification OA with respect to the reduced dimensionality with the Houston dataset: (a) LMNN; (b) ITML; (c) MCML; (d) NCA; (e) AWLML; (f) ML-CC; and (g) CMML.

accuracy when the dimensionality is greater than 20 and maintain a better classification accuracy as the dimensionality increases.

The results of the quantitative evaluation of the seven sets of comparison experiments can be illustrated in more detail in Table V to illustrate the advantages of the proposed SSMD.

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TABLE V
CLASSIFICATION ACCURACIES FOR THE HOUSTON DATASET (DIM = 40)

Class	LMNN		ITML		MCML		NCA		AWLML		ML-CC		CMML		
	-	SSMD	-	SSMD	-	SSMD	-	SSMD	-	SSMD	-	SSMD	-	MD	SSMD
C1	**98.31 ±0.13**	98.13 ±0.18	96.36 ±0.09	**97.69** ±0.14	97.60 ±0.27	**99.47 ±0.42**	97.78 ±0.23	**98.58 ±0.19**	96.27 ±0.30	**96.71 ±0.30**	**98.04 ±0.29**	97.87 ±0.29	98.93 ±0.11	**98.93 ±0.18**	98.84 ±0.07
C2	**98.32 ±0.36**	97.96 ±0.25	95.92 ±0.44	**96.28** ±0.04	97.96 ±0.35	**98.49 ±0.47**	97.78 ±0.17	**98.14 ±0.11**	95.21 ±0.26	**96.72 ±0.21**	97.61 ±0.17	**98.14 ±0.56**	98.67 ±0.13	**98.76 ±0.29**	98.49 ±0.03
C3	**98.09 ±0.15**	97.93 ±0.05	99.84 ±0.70	**99.84** ±0.09	98.09 ±0.51	**99.52 ±0.21**	97.61 ±0.14	**98.72 ±0.33**	99.84 ±0.07	**99.84 ±0.05**	97.61 ±0.11	**98.09 ±0.00**	99.52 ±0.02	99.68 ±0.38	**99.68 ±0.12**
C4	98.12 ±0.10	**98.21 ±0.28**	95.89 ±1.28	**98.03** ±0.27	95.89 ±0.48	**97.68 ±0.27**	**98.21 ±0.35**	97.77 ±0.29	95.35 ±1.00	**97.05 ±0.67**	**98.39 ±0.20**	97.68 ±0.18	**98.39 ±0.27**	98.21 ±0.37	97.41 ±0.27
C5	**98.93 ±0.16**	98.84 ±0.11	**99.28** ±0.34	99.10 ±0.35	98.48 ±0.35	**99.10 ±0.35**	98.57 ±0.22	**98.75 ±0.16**	99.19 ±0.38	**99.46 ±0.32**	**98.75 ±0.35**	98.66 ±0.38	95.79 ±0.37	96.78 ±0.75	**97.40 ±0.54**
C6	**95.55 ±0.06**	94.86 ±0.15	97.95 ±0.06	**99.66** ±0.15	96.92 ±0.80	**97.26 ±1.74**	94.52 ±0.18	**96.23 ±0.59**	97.95 ±0.15	**98.29 ±0.37**	95.89 ±0.15	**95.89 ±0.39**	81.85 ±0.15	81.85 ±1.22	**82.88 ±0.15**
C7	90.62 ±0.65	**92.20 ±0.19**	**93.51** ±0.39	92.38 ±0.63	89.92 ±0.57	**93.34 ±0.83**	85.54 ±0.86	**91.94 ±0.67**	92.02 ±0.64	**93.34 ±0.13**	88.26 ±0.46	**92.02 ±1.53**	91.67 ±0.45	92.38 ±1.73	**93.08 ±0.51**
C8	91.24 ±0.16	**93.21 ±0.07**	87.31 ±1.44	**87.40** ±0.44	92.94 ±1.37	**94.01 ±1.02**	88.56 ±0.61	**93.66 ±0.45**	**86.00 ±0.69**	85.97 ±0.40	92.76 ±0.52	**93.39 ±0.33**	91.78 ±0.54	93.21 ±1.62	**93.48 ±0.67**
C9	86.41 ±1.84	**86.94 ±1.79**	80.99 ±0.87	**86.50** ±1.85	**83.13 ±1.10**	82.06 ±3.23	82.86 ±1.86	**87.83 ±1.79**	83.66 ±1.26	**84.24 ±1.04**	83.30 ±1.06	**88.63 ±0.40**	82.33 ±1.77	85.26 ±0.93	**89.52 ±1.76**
C10	91.76 ±0.83	**92.93 ±0.62**	**94.20** ±2.29	92.93 ±2.06	90.31 ±1.41	**91.67 ±2.00**	88.22 ±1.10	**92.84 ±1.04**	**91.85 ±1.34**	91.76 ±0.64	89.04 ±1.10	**94.93 ±1.79**	86.32 ±1.88	87.50 ±2.37	**91.21 ±1.75**
C11	90.10 ±0.93	**91.36 ±0.83**	86.86 ±2.15	**87.85** ±2.38	**88.93 ±0.87**	88.57 ±1.17	85.15 ±0.46	**92.98 ±2.49**	87.58 ±1.69	**88.93 ±0.95**	88.30 ±1.96	**92.35 ±2.17**	82.90 ±2.46	88.68 ±2.65	**90.46 ±2.17**
C12	90.35 ±0.89	**90.80 ±1.12**	90.35 ±2.21	**92.34** ±2.54	**89.45 ±1.10**	88.10 ±3.92	81.33 ±1.57	**92.79 ±1.65**	89.18 ±0.70	**91.79 ±1.27**	87.02 ±0.91	**92.43 ±1.42**	85.75 ±2.47	88.04 ±2.28	**91.52 ±2.44**
C13	35.07 ±1.16	**37.68 ±1.14**	41.23 ±1.24	**44.79** ±0.74	34.12 ±0.81	**38.15 ±1.87**	26.54 ±1.14	**36.49 ±1.71**	**36.73 ±2.87**	35.55 ±3.46	37.20 ±0.77	**37.20 ±1.49**	38.15 ±0.99	**41.00 ±0.91**	38.15 ±0.69
C14	97.66 ±0.14	**98.96 ±0.00**	99.22 ±0.38	**99.22** ±0.39	98.70 ±2.80	**99.22 ±0.74**	96.62 ±0.11	**98.96 ±0.69**	97.40 ±0.92	**97.40 ±0.62**	98.18 ±0.21	**99.22 ±0.53**	99.22 ±0.42	**99.22 ±0.35**	98.44 ±0.42
C15	**99.49 ±0.05**	99.33 ±0.09	**99.49** ±0.47	99.33 ±0.14	**99.49 ±0.48**	99.16 ±0.09	99.33 ±0.25	**99.49 ±0.07**	99.33 ±0.07	**99.33 ±0.09**	99.33 ±0.12	**99.33 ±0.20**	99.33 ±0.12	99.16 ±0.30	**99.33 ±0.06**
OA	92.24± 0.48	**92.86± 0.35**	91.50 ±0.96	**92.45 ±0.81**	91.48 ±0.63	**92.34 ±1.06**	89.40 ±0.45	**93.29 ±0.77**	90.99 ±0.69	**91.71 ±0.54**	91.25 ±0.43	**93.34 ±0.61**	90.37 ±0.75	91.55 ±0.99	**92.76 ±0.74**
kappa	0.9161 ± 0.0052	**0.9228 ± 0.0039**	0.9081 ± 0.0104	**0.9184 ± 0.0088**	0.9079 ± 0.0069	**0.9172 ± 0.0115**	0.8854 ± 0.0049	**0.9274 ± 0.0084**	0.9025 ± 0.0075	**0.9104 ± 0.0059**	0.9054 ± 0.0047	**0.9280 ± 0.0066**	0.8959 ± 0.0081	0.9086 ± 0.0108	**0.9217 ± 0.0082**

Overall, the new algorithm using SSMD has better classification results and can improve the accuracy by 1% on average, relative to the original algorithm, with NCA improving the accuracy by up to 4%.

As shown in Fig. 13, the new algorithm using SSMD achieves a smoother classification map than the corresponding original algorithm on the Houston dataset. It is clear that the new algorithm achieves the best classification results in most surface cover classes compared to the original algorithm, particularly in small sample areas, such as class 13 (Parking Lot 2) and class 14 (Tennis Court), where the error rate is lowest.

3) Pavia University Dataset: Fig. 14 gives the effect of the size of the subspace dimension on the classification results in different feature reduction algorithms. As seen from Fig. 14, except for the NCA algorithm, the new algorithms in the other six sets of experiments have good classification results when the dimensionality is equal to 5, and they consistently maintain this advantage as the dimensionality grows. In the comparison experiment of the NCA algorithm, the new algorithm also achieves better classification results than the original algorithm for dimensions greater than 15. For the results of the NCA algorithm, we think that after the mapping, in the new feature space, the useful information is concentrated in the first few dimensions, so NCA has a good classification result in low dimensions. As the dimension increases, the redundant information continues to increase, so the accuracy of NCA keeps decreasing.

To make a more detailed comparison of the seven groups of comparison experiments, we present a quantitative presentation of the experimental results in Table VI. From Table VI, it can be seen that the classification accuracy of most classes has been improved with the use of SSMD, such as class 1 (Asphalt), class 2 (Meadows), and class 6 (Bare Soil). Overall, the values of OA and kappa also improved significantly, with OA improving by an average of 2% and the largest improvement reaching 5%.

To further demonstrate the effectiveness of the SSMD, Fig. 15 shows the classification maps based on different algorithms. From the classification plots, we can see that the classification results of the new algorithm using SSMD have a clear advantage on the University of Pavia dataset; for

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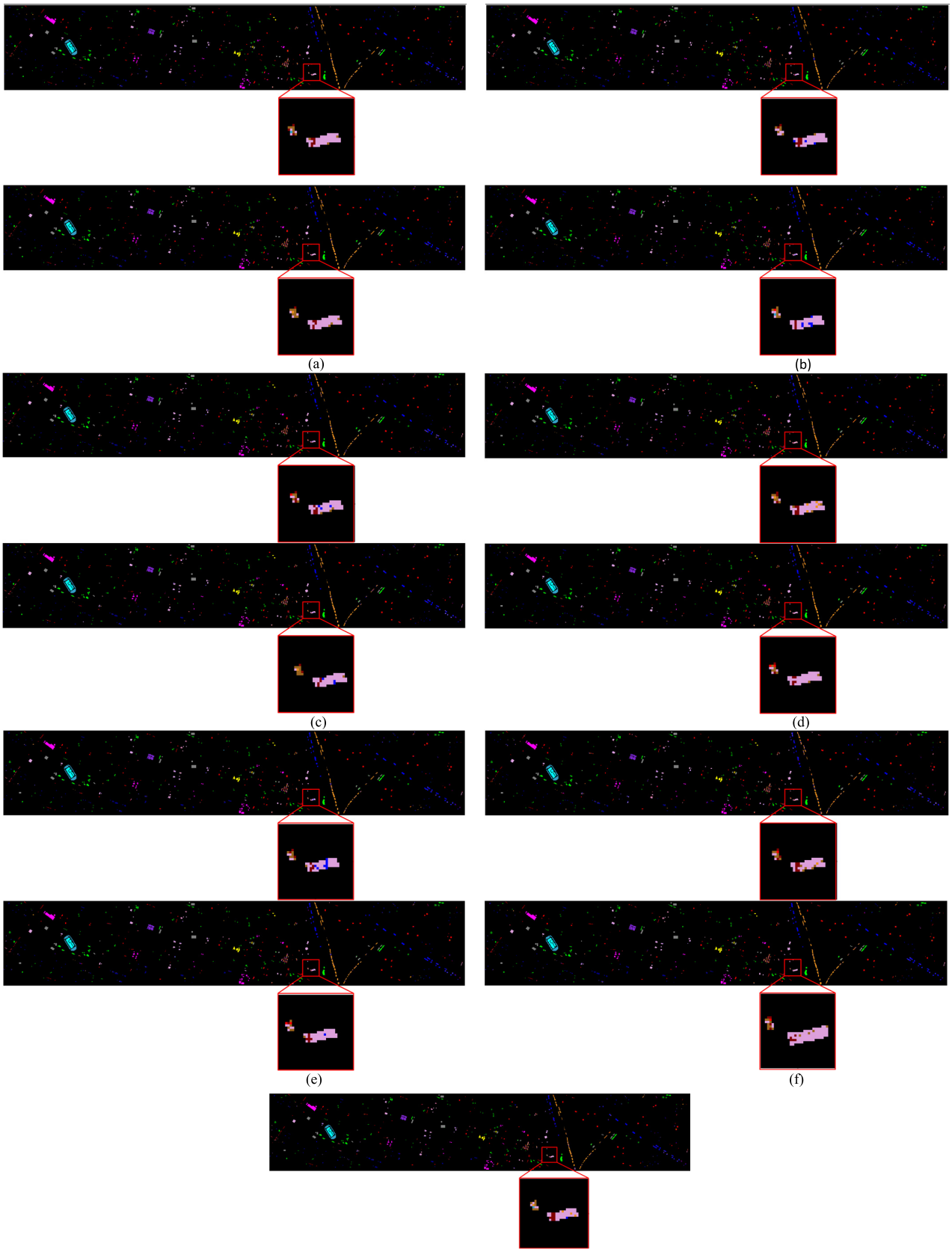


Fig. 13. Classification results for all the methods with the Houston dataset based on 1NN: (a) LMNN; (b) ITML; (c) MCML; (d) NCA; (e) AWLML; (f) ML-CC; and (g) CMML (Top: original and Bottom: new).

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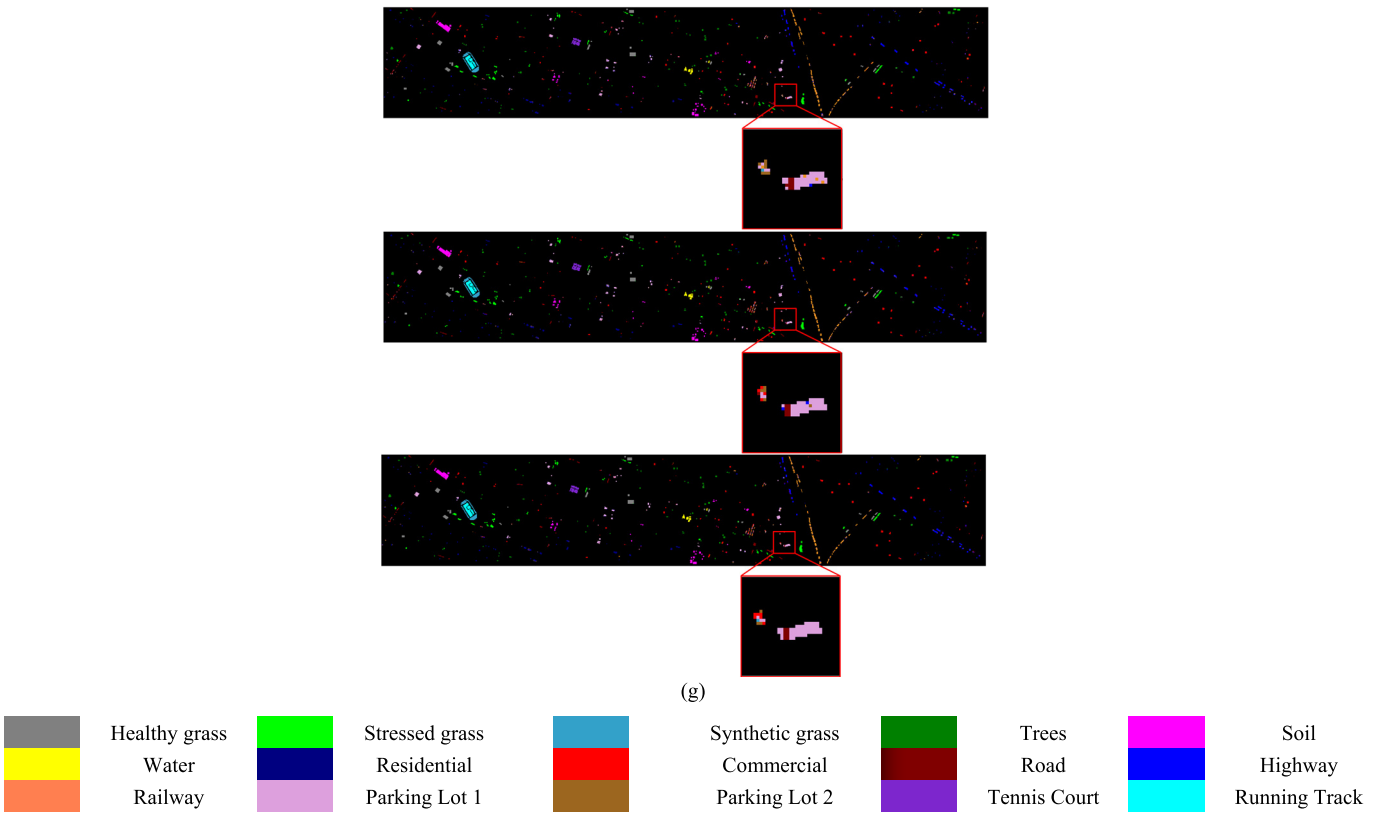


Fig. 13. *(Continued.)* Classification results for all the methods with the Houston dataset based on 1NN: (a) LMNN; (b) ITML; (c) MCML; (d) NCA; (e) AWLML; (f) ML-CC; and (g) CMML (Top: original and Bottom: new).

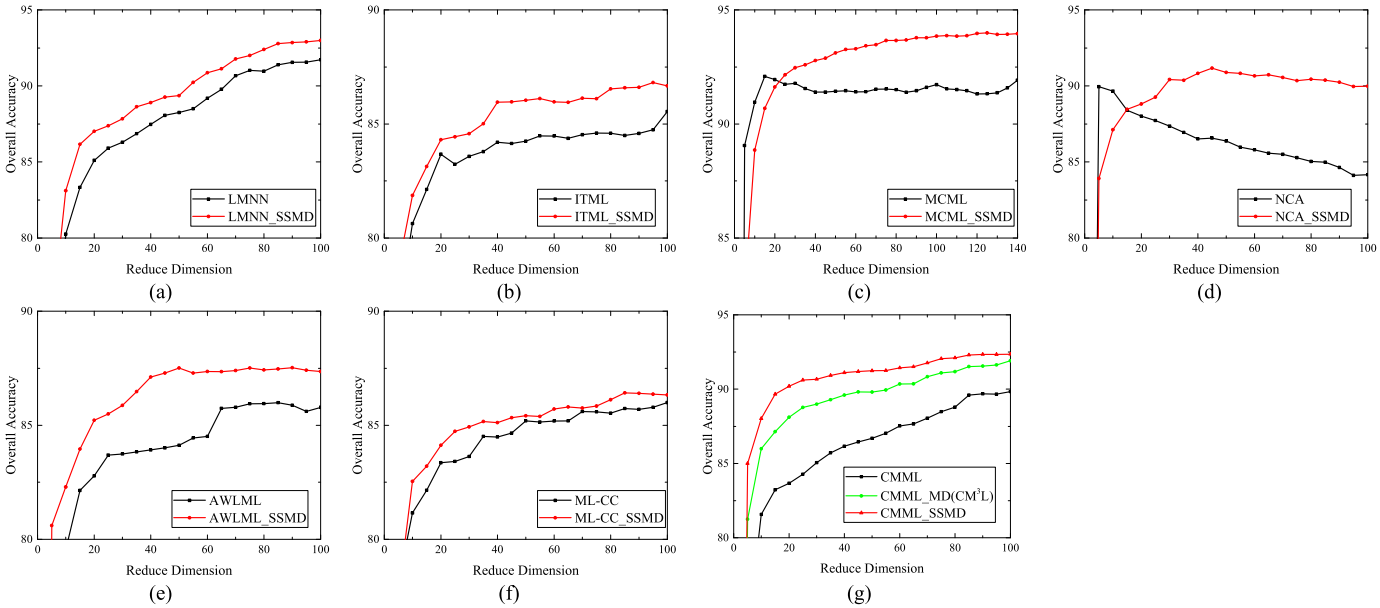


Fig. 14. Classification OA with respect to the reduced dimensionality with the Pavia University dataset: (a) LMNN; (b) ITML; (c) MCML; (d) NCA; (e) AWLML; (f) ML-CC; and (g) CMML.

example, when distinguishing between class 2 (Meadows) and class 6 (Bare Soil), the new algorithm in the NCA, AWLML, and ML-CC experimental group can distinguish well between class 2 (Meadows) and class 6 (Bare Soil), while the original corresponding algorithm tends to confuse the two classes.

D. Analysis of the Running Time

Table VII shows the comparison of the running times of the seven groups of comparison experiments. From Table VII, we can find that the running time of CMML, LMNN, ITML, AWLML, and ML-CC increases after using SSMD. This is

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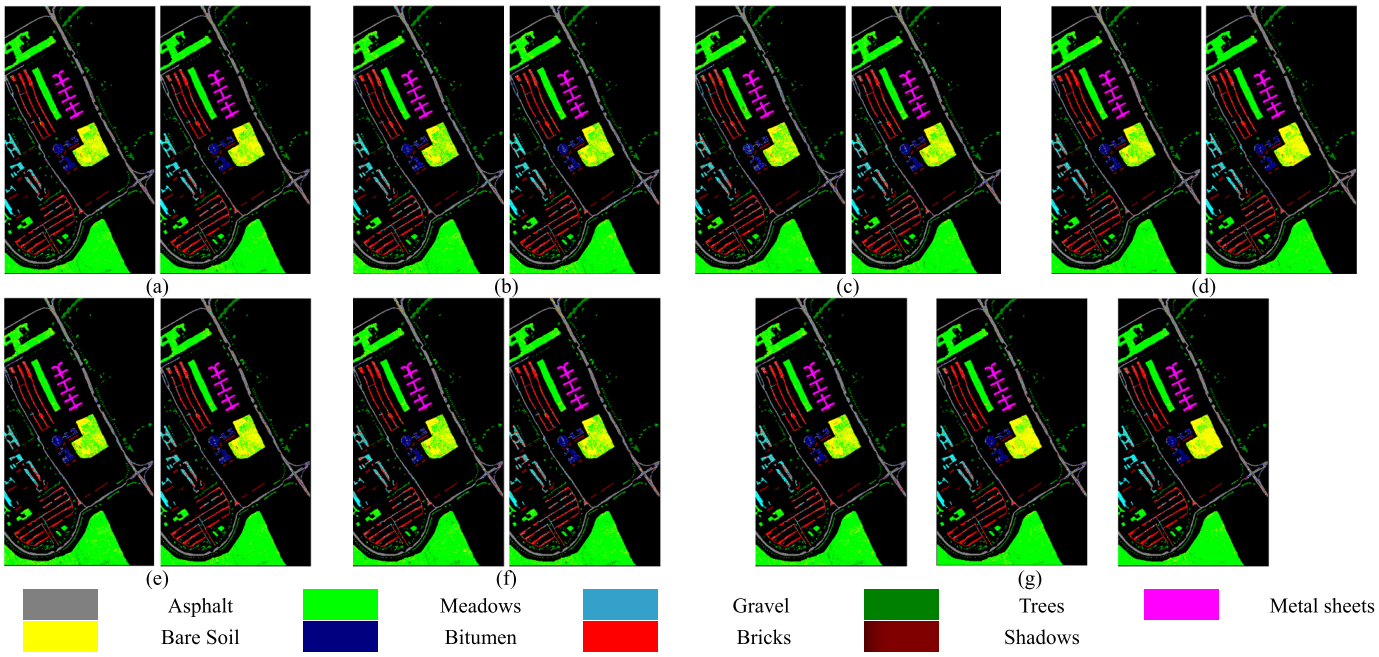


Fig. 15. Classification results for all the methods with the Pavia University dataset based on 1NN: (a) LMNN; (b) ITML; (c) MCML; (d) NCA; (e) AWLML; (f) ML-CC; and (g) CMML (Left: original and Right: new).

TABLE VI

CLASSIFICATION ACCURACIES FOR THE PAVIA UNIVERSITY DATASET (DIM = 40)

Class	LMNN		ITML		MCML		NCA		AWLML		ML-CC		CMML		
	-	SSMD	-	SSMD	-	SSMD	-	SSMD	-	SSMD	-	SSMD	-	MD	SSMD
C1	87.56 ±0.75	**88.45 ±1.01**	85.82 ±1.21	**87.01 ±0.62**	84.85 ±0.33	**87.90 ±0.10**	85.84 ±0.20	**89.69 ±0.06**	83.61 ±1.15	**84.82 ±0.11**	84.23 ±0.52	**85.29 ±0.65**	86.71 ±0.88	85.82 ±0.60	**88.32 ±0.47**
C2	95.00 ±0.72	**95.50 ±0.71**	91.76 ±0.46	**93.71 ±0.36**	92.96 ±0.17	**95.26 ±0.34**	95.31 ±0.11	**97.32 ±0.05**	92.42 ±0.33	**95.41 ±0.27**	**94.27 ±0.26**	93.80 ±0.37	92.37 ±0.97	**96.81 ±0.32**	96.77 ±0.26
C3	54.47 ±2.02	**74.27 ±2.38**	64.16 ±1.09	**66.86 ±1.10**	52.73 ±1.29	**67.13 ±0.83**	61.04 ±0.50	**73.27 ±0.44**	62.41 ±5.04	**64.43 ±0.67**	**56.59 ±0.72**	55.21 ±0.71	69.83 ±1.15	75.28 ±1.56	**76.87 ±1.27**
C4	85.71 ±0.65	**87.01 ±0.62**	**83.93 ±0.75**	83.61 ±0.73	84.91 ±2.30	**85.49 ±0.67**	86.43 ±0.11	**89.70 ±0.14**	81.76 ±0.72	**85.64 ±0.39**	**84.22 ±0.19**	84.08 ±0.47	87.23 ±0.97	92.38 ±0.24	**92.56 ±0.22**
C5	99.26 ±0.04	**99.50 ±0.07**	**99.26 ±0.07**	99.17 ±0.07	99.17 ±0.07	**99.50 ±0.02**	99.42 ±0.03	**99.50 ±0.07**	99.42 ±0.03	**99.50 ±0.04**	**99.59 ±0.06**	99.01 ±0.03	99.50 ±0.03	99.59 ±0.05	**99.59 ±0.03**
C6	57.64 ±4.43	**73.11 ±4.52**	59.19 ±2.07	**64.52 ±2.18**	64.49 ±0.90	**67.15 ±1.54**	63.70 ±0.40	**82.46 ±0.20**	61.91 ±1.15	**72.18 ±0.70**	59.81 ±0.67	**65.67 ±0.59**	66.11 ±3.33	82.21 ±1.64	**83.56 ±1.13**
C7	77.19 ±1.52	**86.22 ±1.19**	**79.53 ±1.50**	78.70 ±1.57	78.03 ±3.23	**82.62 ±0.65**	80.79 ±0.63	**86.47 ±0.39**	73.43 ±3.70	**80.45 ±0.20**	78.20 ±0.56	**80.95 ±0.93**	82.54 ±0.80	85.46 ±0.79	**86.38 ±0.71**
C8	80.38 ±2.06	**81.62 ±1.21**	79.96 ±0.42	**81.74 ±0.43**	76.67 ±1.33	**84.00 ±0.30**	**83.49 ±0.18**	82.40 ±0.16	79.60 ±2.79	**80.68 ±0.55**	**80.11 ±0.26**	79.23 ±2.14	80.65 ±1.79	81.38 ±0.73	**83.07 ±0.68**
C9	**99.88 ±0.05**	99.65 ±0.15	100 ±0.00	**100 ±0.00**	**99.88 ±0.06**	99.06 ±0.06	**99.88 ±0.05**	99.65 ±0.05	**99.88 ±0.10**	99.77 ±0.03	99.65 ±0.07	**99.88 ±0.05**	98.94 ±0.17	99.30 ±0.12	**99.30 ±0.05**
OA	87.80 ±0.86	**88.86 ±1.00**	84.11± 0.49	**86.01 ±0.45**	84.28± 0.63	**87.59 ±0.40**	86.56 ±0.12	**91.16 ±0.10**	83.61 ±0.77	**87.30 ±0.11**	84.66 ±0.25	**85.22 ±0.19**	85.98 ±1.22	90.47 ±0.54	**91.27 ±0.46**
kappa	0.8368 ± 0.0117	**0.8512 ± 0.0135**	0.7876 ± 0.0067	**0.8127 ± 0.0063**	0.7900 ± 0.0089	**0.8336 ± 0.0055**	0.8198 ± 0.0017	**0.8822 ± 0.0014**	0.7809 ± 0.0106	**0.8303 ± 0.0015**	0.7940 ± 0.0033	**0.8024 ± 0.0026**	0.8133 ± 0.0164	0.8733 ± 0.0080	**0.8839 ± 0.0062**

TABLE VII

RUNNING TIME (IN S) ANALYSIS FOR OBTAINING THE DISTANCE METRIC FOR THE THREE DATASETS

Data	LMNN		ITML		MCML		NCA		AWLML		ML-CC		CMML		
	-	SSMD	-	SSMD	-	SSMD	-	SSMD	-	SSMD	-	SSMD	-	MD	SSMD
KSC	33	126	22	115	1010	230	189	17	30	196	773	126	25	49	55
Houston	203	1023	15	81	12405	1046	1207	300	110	251	4080	5638	80	176	183
Pavia	23191	36302	11	45	15230	2248	5039	1367	332	582	7536	10235	275	223	236

because the SSMD is essentially a weighted distance based on the Mahalanobis distance, so the computation time increases compared to the Mahalanobis distance. For the MCML and NCA algorithms, the running time is reduced by using SSMD

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because the fit state can be reached faster and fewer iterations are required after using SSMD. From the seventh set of experiments, it can be seen that using SSMD instead of MD does not have much effect on the runtime.

V. Conclusion

This article proposed a spectral–spatial manifold distance for metric learning, aimed at enhancing the performance of metric learning in HSI DR and classification. The SSMD introduces the idea of manifold learning based on the traditional Mahalanobis distance, combining spatial–spectral information to filter out suitable nearest neighbors in labeled and unlabeled data in order to participate in the construction of manifolds, and then calculates the distance between manifolds and replaces the traditional Mahalanobis distance, thus enhancing the discriminative power by making full use of the neighborhood information provided by the HSI data. To verify the effectiveness of SSMD, a total of seven sets of comparison experiments were constructed in this article. In the seven sets of comparison experiments, seven metric learning algorithms were selected, and seven new algorithms were obtained using SSMD for a comparison with the corresponding original algorithms. The seventh set of experiments added a comparison with CM^3L to compare MD and SSMD. According to the experimental results, it was determined that the measure learning algorithms using SSMD had better experimental results. Based on the experimental results, it was determined that the measurement learning algorithms using SSMD have better experimental results.

However, the SSMD only exploits the information in the unlabeled data at a superficial level; there is still much information in the unlabeled data that is not being exploited. Our future work aims to dig deeper into the unlabeled data to further improve the performance of the SSMD.

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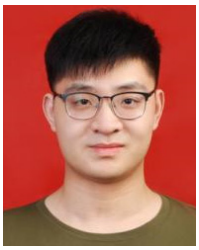
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Yao Jin received the degree in geographic information science from Huazhong Agricultural University, Wuhan, China, in 2018. He is currently pursuing the master's degree in resources and environment with the Hubei Subsurface Multi-Scale Imaging Key Laboratory, Institute of Geophysics and Geomatics, China University of Geosciences, Wuhan.

His research interests include remote sensing and machine learning, especially hyperspectral image dimensionality reduction.



Yanni Dong (Senior Member, IEEE) received the B.S. degree in sciences and techniques of remote sensing from Wuhan University, Wuhan, China, in 2012, and the Ph.D. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, in 2017.

She was a Hong Kong Scholar with the Department of Land Surveying and Geo-Informatics, The Hong Kong Polytechnic University, Hong Kong. She is currently an Associate Professor with the Hubei Subsurface Multi-Scale Imaging Key Laboratory, Institute of Geophysics and Geomatics, China University of Geosciences, Wuhan. Her research interests include hyperspectral image processing, pattern recognition (PR), and machine learning.

Dr. Dong serves as a reviewer for more than 20 international journals, including the IEEE TRANSACTIONS ON CYBERNETICS (TCYB), IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS), IEEE TRANSACTIONS ON MULTIMEDIA (TMM), IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS), IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL), and PR. She regularly serves as a PC Member of IJCAI and AAAI.



Yuxiang Zhang (Member, IEEE) received the B.S. degree in sciences and techniques of remote sensing and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2011 and 2016, respectively.

She was a Visiting Scholar with The University of Sydney, Sydney, NSW, Australia, from 2019 to 2020. She is currently an Associate Professor with the Institute of Geophysics and Geomatics, China University of Geosciences, Wuhan. Her research interests include hyperspectral image processing, pattern recognition (PR), and machine learning.



Xiangyun Hu received the B.S. and M.Sc. degrees in geophysics from the Changchun College of Geology, Changchun, China, in 1988 and 1991, respectively, and the Ph.D. degree in mineral resource prospecting and exploration from the China University of Geosciences, Wuhan, China, in 2000.

Since 2005, he has been serving as a Professor with the Institute of Geophysics and Geomatics, China University of Geosciences. His research interests include 3-D magnetotellurics (MT)/controlled-source audio-frequency MT (CSAMT)/controlled-source electromagnetics (CSEM) forward modeling and inverse problems, geoelectrical characteristics of the Tibetan Plateau, and integrated geophysical interpretation.

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